
Ultra-Low Precision 4-bit Training of Deep Neural Networks

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Abstract

In this paper, we propose a number of novel techniques and numerical representation formats that enable, for the very first time, the precision of training systems to be aggressively scaled from 8-bits to 4-bits. To enable this advance, we explore a novel adaptive Gradient Scaling technique (GradScale) that addresses the challenges of insufficient range and resolution in quantized gradients as well as explores the impact of quantization errors observed during model training. We theoretically analyze the role of bias in gradient quantization and propose solutions that mitigate the impact of this bias on model convergence. Finally, we examine our techniques on a spectrum of deep learning models in computer vision, speech and NLP. In combination with previously proposed solutions for 4-bit quantization of weight and activation tensors, 4-bit training shows non-significant loss in accuracy across application domains while enabling significant hardware acceleration ($>7\times$ over state of the art FP16 systems).

1 Introduction

Over the past decade, Deep Neural Networks (DNNs) have surpassed traditional Machine Learning techniques on a wide spectrum of application domains including speech [1–3], computer vision [4–10], and language [11, 12]. As models and datasets have grown in complexity, DNN training times have increased significantly—limiting the pace of innovation in model training and deployment life cycles.

Driven by a number of key advances in low-precision arithmetic [13–16] and the quadratic dependency of throughput on precision, reduced-precision training has become the de facto technique to boost the performance and power efficiency of deep learning hardware. 16-bit floating point training formats have been integrated into multiple accelerator products [17–19] and shown to provide 4–8 times the performance of 32-bit hardware designs. Recently, new insights into low-precision accumulation errors, batchnorm statistics and novel 8-bit floating point formats have resulted in successful demonstrations of 8-bit training for a wide range of deep learning tasks—resulting in a further $>2\text{--}4\times$ boost [15] in training system performance over 16-bit systems.

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